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ECG-Based Arrhythmia Classification: A Comparative Study of Classical Machine Learning and Deep Learning Architectures Using Metaheuristic Hyperparameter Optimization

Deep Learning Mini Project – Phase 3 (Final Report)
[ICT-4442]

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Code repository: https://github.com/CHINMAYBHT/Arrhythmia-Detection

Abstract

Automatic heartbeat classification from the electrocardiogram (ECG) could make continuous cardiac monitoring practical. Many published accuracies, however, come from train/test splits where beats from the same patient end up on both sides. In this project we put three model families through a single patient-independent protocol on the MIT-BIH Arrhythmia Database, using the DS1/DS2 record split and the five AAMI EC57 beat classes. The first is an SVM and a Random Forest trained on hand-crafted morphology, wavelet and RR-interval features. The second is a 1D convolutional neural network (CNN) that reads the raw beat window. The third is a CNN-LSTM that looks at the five beats ending at the current one before making its call. Alongside this, we checked whether two metaheuristic optimizers, the Grey Wolf Optimizer (GWO) and the Whale Optimization Algorithm (WOA), tune the CNN's hyperparameters better than a grid search given the same budget. On the 49,697 held-out test beats, the Random Forest came out on top in both accuracy (0.860) and macro-F1 (0.393), with the SVM next (macro-F1 0.353). The best deep model, a WOA-tuned CNN, managed 0.346, and the CNN-LSTM 0.335. Every model picks up ventricular ectopic beats reasonably well (best F1 0.79). None of them reliably catches supraventricular (S) or fusion (F) beats, though, which keeps macro-F1 between 0.32 and 0.39. GWO and WOA did find configurations that scored higher on validation than anything the grid search found, but the advantage vanished on the test patients: the best tuned CNN (WOA) beat the untuned CNN on macro-F1 by just 0.005. Under a strict inter-patient protocol, then, the model family and the tuner matter much less than the small pool of training patients and the heavy class imbalance.

Keywords: ECG, arrhythmia classification, MIT-BIH, AAMI EC57, inter-patient evaluation, 1D-CNN, CNN-LSTM, Random Forest, SVM, Grey Wolf Optimizer, Whale Optimization Algorithm.

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1 Introduction

Cardiac arrhythmias are disturbances of the heart's rhythm, and some of them are early warnings of serious cardiovascular disease. Normally a cardiologist spots them by reading an ECG trace. That works, but it is slow, it varies from one reader to the next, and it cannot keep up with the hours of signal that Holter monitors and wearable devices now record. Automatic beat classification has therefore been an active research topic for a long time, and the MIT-BIH Arrhythmia Database [1], [2] is the benchmark almost everyone reports on.

Broadly, the literature splits into two camps. Classical methods compute hand-crafted features for each beat (RR intervals, QRS morphology, wavelet coefficients and so on) and train a classifier such as an SVM or a Random Forest on top of them [12]. Deep learning methods skip the feature engineering and learn straight from the raw signal, using 1D CNNs [3], [13], [14] or recurrent and hybrid CNN-LSTM networks that also model how one beat follows another [5], [7], [15]. Many of the headline accuracies above 95%, however, come from intra-patient splits, where the same patient's beats sit in both the training and the test set. Switch to an inter-patient split, so that test patients are never seen during training, and performance on the minority classes falls sharply [4], [5], [12], [16].

Our project takes one model from each camp and runs them all on the same data, with the same patient-wise split and the same metrics, so that whatever differences show up can be put down to the models themselves. On the side, we ask whether two nature-inspired metaheuristic optimizers, GWO [9] and WOA [10], [17], can find better CNN hyperparameters than a grid search with the same compute budget. So far, that question has mostly been looked at for medical images [11].

2 Literature Review

The work most relevant to us falls into four groups: classical feature-based classifiers, convolutional networks, recurrent and hybrid networks, and metaheuristic hyperparameter optimization. Table 1 summarises the main papers.

2.1 Evaluation protocol and classical feature-based methods

De Chazal, O'Dwyer and Reilly [12] introduced the inter-patient DS1/DS2 split of MIT-BIH that we adopt here. They showed that a linear discriminant classifier on morphology and RR-interval features is highly sensitive to V beats in unseen patients but does much worse on S beats. Their work also made RR intervals essential for the S class, because a supraventricular beat looks almost normal and differs mainly in arriving early. The five beat classes we use (N, S, V, F, Q) come from the AAMI EC57 standard [18]. Ben Slama et al. [8] found that an SVM trained on CNN-extracted features can beat the CNN on its own (97.33% against 85.52%). A classical classifier given good features can therefore outperform a deep network. Nikandish, He and Haghi [6] fused wavelet, interval and area features and lifted the accuracy on difficult beat types considerably. Together these papers shaped Model 1: an SVM [19] and a Random Forest [20] on morphology, wavelet and RR-interval features. Finally, Silva et al. [16] reviewed studies from 2017 to 2024 and found that few of them follow the inter-patient protocol, which makes

published numbers hard to compare. That is part of why we hold every model to one common protocol.

2.2 Convolutional networks

Kiranyaz, Ince and Gabbouj [13] showed that a compact 1D-CNN can learn its own features from raw ECG beats, though their best results relied on patient-specific training. Acharya et al. [14] trained a 9-layer CNN on MIT-BIH beats and reported 94% accuracy, but on an intra-patient split. Khan et al. [3] reached 95.2% accuracy across five heartbeat classes without saying whether their split kept patients apart. Guo, Sim and Matuszewski [4] tested DenseNet and GRU models on the inter-patient split and got an F1 of 61.25 for S beats against 89.75 for V beats. Even for deep networks, then, S is the hardest class. Model 2 is a 1D-CNN along these lines.

2.3 Recurrent and hybrid networks

LSTM networks [15] are built to capture long-range dependencies in sequences. Yildirim [7] paired a wavelet sequence with a bidirectional LSTM for ECG classification. Alamatsaz et al. [5] proposed a lightweight 11-layer CNN-LSTM with 98.24% mean accuracy across several databases. Mousavi, Afghah and Acharya [21] combined a CNN encoder with a sequence-to-sequence decoder; by modelling beat sequences and dealing with class imbalance head-on, they reported a high inter-patient S-class sensitivity of 88.94%. Model 3 borrows that idea. A shared CNN encodes each beat, and an LSTM reads the sequence of five beats.

2.4 Metaheuristic hyperparameter optimization

The Grey Wolf Optimizer [9] imitates the leadership hierarchy and hunting behaviour of grey wolves, and the Whale Optimization Algorithm [10] imitates the bubble-net hunting of humpback whales; Mohammed, Umar and Rashid [17] survey its many variants. Hamad, Samma and Suandi [11] reviewed metaheuristic CNN tuning for medical image diagnosis between 2019 and 2022. GWO and WOA came up often, but almost every study dealt with images, and about 63.6% were about COVID-19 detection. We could not find any study that tunes an inter-patient ECG beat classifier with GWO or WOA and compares it against a grid search with an equal budget, so our side experiment does that.

Table 1: Summary of the main reviewed papers.

Ref	Paper	Method	Dataset	Key result	Relevance
[12]	de Chazal et al. (2004)	LDA on morphology + RR features	MIT-BIH, inter-patient DS1/DS2	First standard inter-patient protocol; S class much harder than V	Our split; RR features (Model 1)
[4]	Guo et al. (2018)	DenseNet + GRU	MIT-BIH, inter-patient	F1 61.25 (S), 89.75 (V)	S is hardest even for deep nets
[3]	Khan et al. (2020)	CNN, 5 classes	MIT-BIH	95.2% accuracy (split not stated)	Reference for Model 2
[13]	Kiranyaz et al. (2016)	Compact 1D-CNN on raw beats	MIT-BIH	Learns features from raw ECG; patient-specific training	Basis of Model 2
[14]	Acharya et al. (2017)	9-layer CNN	MIT-BIH (intra-patient)	94% accuracy	Deep CNN baseline
[21]	Mousavi et al. (2018)	CNN + seq2seq	MIT-BIH, AAMI EC57	Inter-patient S sensitivity 88.94%	Beat sequences + imbalance (Model 3)
[5]	Alamatsaz et al. (2022)	Lightweight CNN-LSTM	MIT-BIH, LTAF	98.24% mean accuracy	Closest to Model 3
[7]	Yildirim (2018)	Wavelet sequence + Bi-LSTM	MIT-BIH	High accuracy on 5 rhythm classes	Recurrent ECG modelling
[6]	Nikandish et al. (2024)	Feature fusion + (Bi-)LSTM	MIT-BIH	RBBB 31.4% → 84.3% with area features	Wavelet / interval features
[8]	Ben Slama et al. (2025)	CNN features + SVM	DeepQ	CNN 85.52%, CNN-SVM 97.33%	Classical vs deep (Model 1)
[9], [10]	Mirjalili et al. (2014, 2016)	GWO, WOA	Benchmark functions	Competitive with PSO, GSA, DE	Our two optimizers
[11]	Hamad et al. (2024)	Review of metaheuristic CNN tuning	Medical images	GWO-tuned CNN 97.78% (COVID X-ray)	Motivation; no ECG studies
[16]	Silva et al. (2025)	Systematic review	ECG studies 2017–2024	Few use inter-patient splits	Justifies common protocol

3 Objectives

1. Build one shared, patient-independent preprocessing pipeline for MIT-BIH so that every model gets exactly the same inputs and test data.
2. Implement a classical baseline (SVM / Random Forest), a 1D-CNN and a CNN-LSTM, compare them on the same metrics (accuracy, per-class precision/recall/F1, macro-F1, confusion matrix), and work out why they perform differently.
3. Find out whether tuning the 1D-CNN's hyperparameters with GWO and WOA helps, compared with both the untuned network and a grid search on the same compute budget.

4 Methodology

4.1 Shared data pipeline

Every model draws on the MIT-BIH Arrhythmia Database [1]: 48 half-hour, two-channel ambulatory recordings from 47 subjects, sampled at 360 Hz, with each beat annotated by cardiologists. We download the data with the `wfdb` library (`src/download_data.py`) and process it once with `src/preprocessing.py`:

1. **Filtering.** We band-pass filter channel 0 (MLII) between 0.5 and 40 Hz using a third-order zero-phase Butterworth filter, which removes baseline wander and high-frequency noise.
2. **Segmentation.** Around every annotated beat we cut a 200-sample window: 90 samples before the R-peak and 110 after it, or about 0.56 s.

3. **Labelling.** MIT-BIH symbols are mapped onto the AAMI EC57 classes [18]: N (N, L, R, e, j), S (A, a, J, S), V (V, E), F (F) and Q (/, f, Q). Anything that is not a beat annotation is thrown away.
4. **RR intervals.** For each beat we compute the previous RR interval, the next RR interval, the mean of the previous ten RR intervals and the ratio of previous to next RR. All of this happens within the beat's own record, so no interval ever spans two recordings.
5. **Patient-independent split.** We follow de Chazal et al. [12]: 22 records make up the training set (DS1) and a different 22 make up the test set (DS2). The four paced records (102, 104, 107, 217) are left out. No patient shows up in both sets.

That leaves 51,008 training beats and 49,697 test beats (Table 2). The imbalance is severe. Roughly 89% of beats are Normal, and the Q class has just 8 training beats and 7 test beats.

Table 2: Class distribution of the patient-independent split.

Class	Train beats	Train %	Test beats	Test %
N – Normal	45,854	89.9%	44,245	89.0%
S – Supraventricular ectopic	944	1.9%	1,837	3.7%
V – Ventricular ectopic	3,788	7.4%	3,220	6.5%
F – Fusion	414	0.8%	388	0.8%
Q – Unknown	8	0.0%	7	0.0%
Total	51,008	100%	49,697	100%

4.2 Model 1: SVM and Random Forest on hand-crafted features (Tanush Ralhan)

We describe each beat with 20 features. Eight are morphology statistics of the window: maximum, minimum, mean, standard deviation, skewness, kurtosis, peak-to-peak amplitude and the sum of absolute first differences. Another eight come from wavelets, namely the mean and standard deviation of each sub-band of a three-level Daubechies-4 decomposition. The last four are the RR-interval features. Standardisation uses statistics from the training set only.

To deal with the imbalance, we oversample the training set with SMOTE [22], bringing each minority class up to 20,000 samples (synthetic plus real) and leaving the N class as it is. The test set is never resampled. On top of that, both classifiers use balanced class weights. For the Random Forest [20] we grid-search the number of trees (200, 400), the maximum depth (unlimited, 20) and the minimum leaf size (1, 2), using three-fold stratified cross-validation scored on macro-F1. The RBF-kernel SVM [19] is tuned over $C \in \{1, 10\}$ and $\gamma \in \{\text{scale}, 0.01\}$ on a 20,000-sample subset. The final SVM is then trained on a 60,000-sample subset of the oversampled data, since SVM training time grows faster than linearly with the number of samples. The search picked 400 trees, unlimited depth and a minimum leaf size of 1 for the Random Forest, and $C = 10$, $\gamma = \text{scale}$ for the SVM. Cross-validated macro-F1 on the oversampled training data came out at 0.997 and 0.984 respectively, far above the test results; Section 6 explains the gap.

4.3 Model 2: 1D-CNN on raw beats (Pranshu Singh)

Each 200-sample window is z-score normalised per beat and passed to a 1D-CNN made of three blocks of Conv1D, batch normalisation, ReLU and max-pooling (32, 64 and 128 filters, kernel size 7, pool size 2). After that come dropout (0.3), global average pooling, a 64-unit dense layer and a five-way softmax, for 81,605 parameters in total. We train it with Adam [23] at a

learning rate of 10^{-3} and a batch size of 128, with balanced class weights in the cross-entropy loss, early stopping on validation loss (patience 8), learning-rate halving on plateau, and a cap of 50 epochs.

4.4 Model 2 tuning: grid search, GWO and WOA (Pranshu Singh)

All three tuners get exactly the same budget: 36 short training runs (no more than 8 epochs each, early-stopping patience 3). Every run is scored by macro-F1 on a validation set holding 15% of the training beats, drawn as a stratified random split; the test set plays no part in the search. Each tuner's best configuration is then retrained on the full training set for up to 50 epochs and evaluated on the test set once.

- **Grid search** covers the learning rate $\{3\times10^{-4}, 10^{-3}, 3\times10^{-3}\}$, the base filter count {16, 32, 64} (the blocks use base, 2× and 4× filters) and the kernel size {3, 5, 7, 9}, giving 3×3×4 = 36 points. Dropout (0.3) and dense units (64) are held fixed.
- **GWO** [9] and **WOA** [10] search a continuous five-dimensional space: $\log_{10}$ learning rate in [-4, -2], dropout in [0.1, 0.5], base filters in [16, 128], dense units in [32, 256] and kernel size in [3, 11] (rounded to an odd integer). Each optimizer runs 6 agents for 6 iterations, which comes to 36 evaluations. In GWO, each wolf moves towards the average of the positions suggested by the three best wolves (alpha, beta and delta), while a coefficient a falls linearly from 2 to 0 so that the search gradually shifts from exploration to exploitation. In WOA, each whale picks one of three moves at random, with equal probability: it encircles the best whale, heads towards a random whale (exploration), or follows a logarithmic spiral around the best whale.

4.5 Model 3: CNN-LSTM hybrid (Chinmai)

The CNN-LSTM classifies each beat from the sequence of five consecutive beats ending at it, which gives it rhythm context that a single-beat model never sees. Sequences are built inside each record (never across two recordings) with a stride of one beat. The first four beats of every record lack a full history and are dropped, which leaves 49,609 test sequences. A shared convolutional encoder (two blocks of Conv1D with 32 and 64 filters, kernel size 5, batch normalisation, ReLU and pooling, then global average pooling) is applied to every beat through TimeDistributed layers. An LSTM with 64 units [15] then reads the five beat vectors, followed by dropout (0.3), a 64-unit dense layer and a softmax over the five classes, for 48,389 parameters. Training uses the same optimiser, class weights and callbacks as the CNN, but with a batch size of 64.

5 Experimental Setup

- **Hardware and software.** Model 1 ran on a laptop CPU with Python 3.14, scikit-learn and imbalanced-learn. TensorFlow has no stable release for Python 3.14 and we had no local GPU, so Models 2 and 3 and all of the tuning ran in Kaggle notebooks on an NVIDIA Tesla T4 GPU (`kaggle_pipeline.ipynb`, `kaggle_phase2.ipynb`). One epoch took about 3 s for the CNN and about 18 s for the CNN-LSTM, and the grid search took 959 s. The final SVM/Random Forest run (feature extraction, SMOTE, both grid searches and the final fits) took around 35 minutes on the CPU.

- **Metrics.** We report accuracy, per-class precision, recall and F1, macro- and weighted-average F1, and the confusion matrix (all from scikit-learn). Since N makes up 89% of the test set, macro-F1 is our main metric for comparison. Q has only 7 test beats, so we also give macro-F1 over N, S, V and F.
- **Common protocol.** Every model is trained on DS1 and evaluated once on DS2. Per-model results live in `results/`, and `src/compare_results.py` rebuilds the comparison table, the error analysis and the figures from them.
- **Seeds.** Each configuration was run once, with the seed fixed at 42.

6 Results and Discussion

6.1 Comparison across models

Table 3 puts all the models side by side on the test set, and Figure 1 plots the per-class F1.

Table 3: Test-set results on the patient-independent DS2 split (single run each). Macro-F1* excludes the Q class (7 test beats).

Model	Acc.	Macro F1	Macro F1*	Wtd F1	F1 N	F1 S	F1 V	F1 F	F1 Q
SVM	0.777	0.353	0.442	0.828	0.872	0.208	0.678	0.009	0.000
Random Forest	**0.860**	**0.393**	**0.491**	0.881	0.924	0.192	0.785	0.064	0.000
1D-CNN (default)	0.727	0.342	0.426	0.800	0.839	0.085	0.779	0.002	0.004
1D-CNN (grid-search tuned)	0.703	0.339	0.422	0.783	0.819	0.090	0.777	0.004	0.003
1D-CNN (GWO tuned)	0.751	0.323	0.403	0.807	0.853	0.053	0.702	0.002	0.003
1D-CNN (WOA tuned)	0.763	0.346	0.432	0.824	0.865	0.070	0.794	0.002	0.003
CNN-LSTM	0.834	0.335	0.419	0.856	0.907	0.049	0.714	0.005	0.000

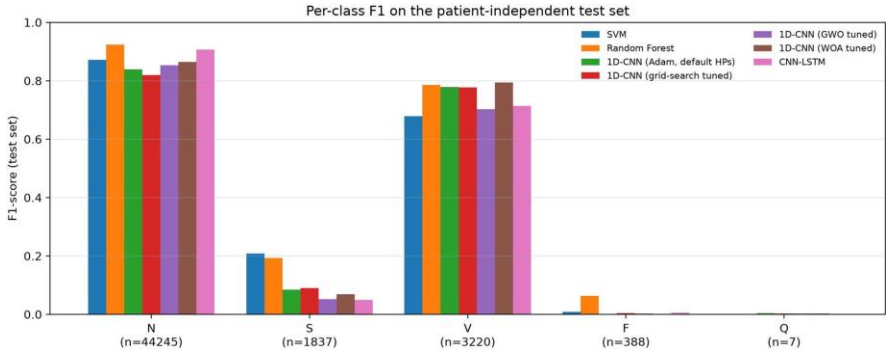


Figure 1: Per-class F1 of every model on the test set.

Of all the models, the Random Forest on hand-crafted features has the best macro-F1 (0.393, or 0.491 without Q), ahead of the best deep model, the WOA-tuned CNN at 0.346. It is the only model that picks up a useful share of S beats (F1 0.19, while no network gets past 0.09), and it also has the best F-class result (F1 0.06, recall 0.29). This comes from the RR-interval features: an S beat looks almost normal but comes early, and the previous-RR and RR-ratio features measure that timing. A network that only sees one 0.56 s window has no way to measure it. These

features were missing from the interim Random Forest run (macro-F1 0.31, S F1 0.00), so most of the improvement since then comes from them and from SMOTE. The SVM, which uses the same features, has the highest S-class F1 (0.21) and the second-best macro-F1 (0.353). However, it labels 5,969 Normal beats as Fusion, and that drags its accuracy down to 0.777. Both classical models also overfit in the same way as the tuning experiment described below. Their cross-validated macro-F1 on the training data was 0.98–1.00, compared with 0.35–0.39 on the test patients. This happens because the cross-validation folds contain beats from the same patients, and SMOTE builds synthetic beats from neighbours that end up in different folds.

The deep models learn the shape of ventricular beats well with no hand-crafted features at all: the CNN variants reach a V-class F1 of 0.70–0.79, on par with the Random Forest (0.79). A ventricular beat has a wide, abnormally shaped QRS complex, and convolutional filters pick that up straight from the waveform. Among the deep models, the CNN-LSTM has the highest accuracy (0.834) and the fewest false alarms on Normal beats (N recall 0.871, against 0.758 for the CNN). Seeing the four preceding beats helps it confirm that a beat is part of a regular normal rhythm. Even so, its macro-F1 (0.335) is no better than that of the single-beat CNN (0.342). What it gains on N it gives back elsewhere, with more Normal beats wrongly labelled S (3,871) and a lower V precision (0.62). The LSTM cannot learn RR timing from the beat sequence either, because each input window is centred on its own R-peak, so the gaps between beats never show up in the sequence of windows; only the shapes of the neighbouring beats do.

6.2 Error analysis

Figure 2 shows the row-normalised confusion matrices (the raw counts are in `results/error_analysis.txt`). Three error patterns turn up in every model.

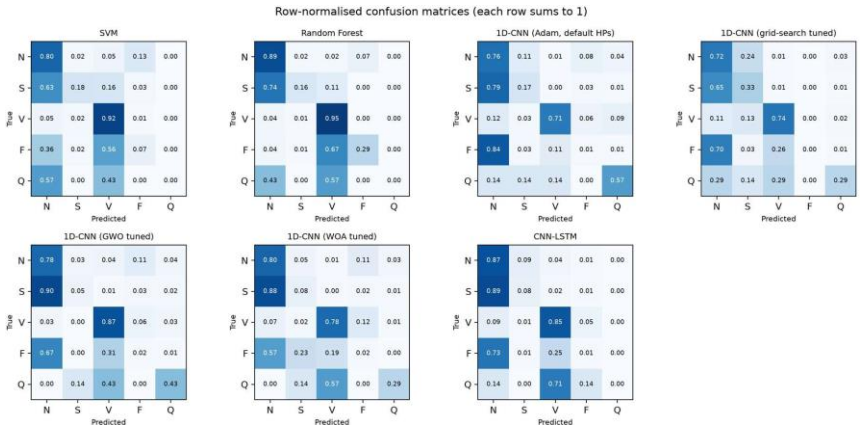


Figure 2: Row-normalised confusion matrices (rows are true classes, each row sums to 1).

1. **S beats are classified as Normal.** Every model predicts most true S beats as N, from 63% for the SVM up to 90% for the GWO-tuned CNN. A supraventricular ectopic beat starts above the ventricles, so its QRS complex looks almost exactly like a normal one; what sets it apart is that it arrives early. The networks only ever see R-peak-centred windows, so this timing is invisible to them, and they miss 65–90% of S beats. The SVM and Random Forest

get the RR intervals and miss the fewest (63% and 74%). Even with timing information, though, they still miss most S beats, because whether a beat is premature has to be judged against the patient's own heart rate, and the training set holds only 944 S beats. The S class is also concentrated in very few patients. Of the 1,837 test S beats, 1,382 (75%) come from a single record (232), and 383 of the 944 training S beats come from record 209. In effect, the models learn the S beats of a handful of training patients and are then tested mostly on one different patient whose beats they fail to generalise to.

2. **Class weights cause many false alarms on Normal beats.** With balanced class weights, a single Q training beat counts roughly 1,275 times as much as an N beat. The networks respond by pushing part of the huge N class into the rare classes. The untuned CNN, for example, labels 5,013 N beats as S, 3,659 as F and 1,733 as Q, and together these account for 77% of its mistakes. Because N is so large, getting even a few percent of N beats wrong is enough to push the precision of S, F and Q close to zero.
3. **Fusion beats are confused with V and N.** A fusion beat is a blend of a normal and a ventricular beat, so its shape falls somewhere between the two. The classical models call most F beats V (56–67%), whereas the networks call most of them N (57–84%), and F-class F1 is at most 0.013 for every model. It does not help that 372 of the 414 training F beats come from one patient (record 208) and 362 of the 388 test F beats from another (record 213), so the F class is learned from one patient and tested on a different one.

The Q-class scores mean very little. With only 7 test beats, a single correct prediction moves Q recall by 0.14.

6.3 Metaheuristic versus grid-search tuning

Table 4 and Figure 3 set the four CNN variants against each other.

Table 4: 1D-CNN tuning experiment. All tuners had the same budget of 36 short training runs.

Variant	Val. macro-F1	Test acc.	Test macro-F1	Selected hyperparameters
Default (Adam)	–	0.727	0.342	lr 10^{-3}, filters 32/64/128, kernel 7, dropout 0.3, dense 64
Grid search	0.462	0.703	0.339	lr 10^{-3}, filters 32/64/128, kernel 7, dropout 0.3, dense 64
GWO	0.627	0.751	0.323	lr 5.5×10^{-4}, filters 78/156/312, kernel 11, dropout 0.16, dense 161
WOA	0.556	0.763	0.346	lr 1.3×10^{-4}, filters 125/250/500, kernel 7, dropout 0.17, dense 75

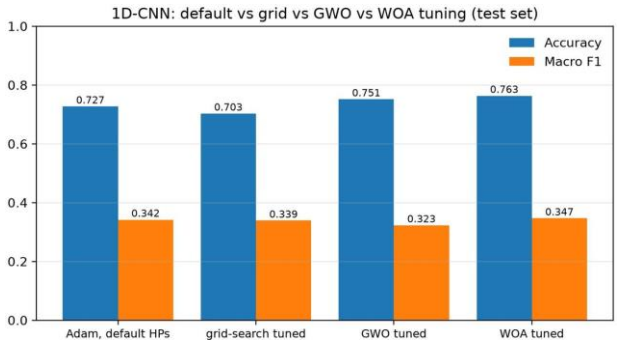


Figure 3: Test accuracy and macro-F1 of the four CNN variants.

Both GWO and WOA found configurations with clearly higher validation macro-F1 than the best grid point (0.627 and 0.556, against 0.462). That is because their continuous search space also takes in dropout, the dense layer size and larger networks, all of which the grid kept fixed. None of this carried over to the test set. WOA lifted test macro-F1 by only about 0.005 over the untuned CNN, and GWO was actually 0.019 worse, even though both raised accuracy by 2–4 points simply by predicting fewer S beats. The grid search ended up selecting exactly the default configuration, yet its retrained model scored 0.024 lower in accuracy than the default run. That alone shows how much a single training run can vary.

The cause is the validation set used during the search. It is a random 15% of the training beats, so it contains beats from the same patients as the training data, while the test set is made up of new patients. The optimizers therefore end up choosing hyperparameters that suit the training patients. Both picked larger networks than the default (78/156/312 filters for GWO and 125/250/500 for WOA, against 32/64/128) and lower dropout, around 0.16-0.17 instead of 0.3. This is the kind of overfitting the inter-patient protocol is meant to expose. On the test set, the spread between configurations (0.323–0.346) is smaller than the variation between two runs of the same configuration. On this dataset, metaheuristic tuning did not pay for itself: each optimizer cost 36 extra training runs and delivered no reliable improvement. A fair test of these optimizers would need a validation set drawn from separate training patients.

6.4 Comparison with published work

Our scores fall well short of the 94–98% accuracies reported in [3], [5], [14], but those studies either use intra-patient splits or do not say which split they used. Under the inter-patient protocol, published S-class F1 values sit around 0.5–0.6 [4], [12], while our models reach 0.09 at best. Most of that gap comes down to techniques we did not use: per-patient RR-interval normalisation [12], sequence-to-sequence modelling with targeted resampling [21], or larger models trained on additional databases [4].

6.5 Limitations

1. **Early stopping on the test set.** The CNN, its tuned retrains and the CNN-LSTM all used the test set as the validation set for early-stopping and learning-rate reduction, so their test scores are slightly optimistic. The SVM and Random Forest are unaffected. Fixing this

means holding out separate training patients for validation and rerunning the networks on a GPU.

2. **Single run per model.** Differences of a few points between models fall within the run-to-run variation (the default CNN, for instance, gave 0.727 and 0.733 accuracy over two runs), so they should not be treated as significant.
3. **The CNN-LSTM is scored on 88 fewer test beats** (the first four beats of each record), so its test set is not quite the same as the other models'.
4. **Q class.** With only 8 training and 7 test beats, Q can be neither learned nor evaluated, which is why we also report macro-F1 without it.

7 Conclusion

We compared an SVM, a Random Forest, a 1D-CNN and a CNN-LSTM for five-class ECG beat classification on MIT-BIH, all under the same patient-independent protocol, and we tested GWO and WOA against a grid search for tuning the CNN. The Random Forest, with RR-interval features and SMOTE, gave the best accuracy (0.860) and macro-F1 (0.393). The two classical models were also the only ones to catch a meaningful share of supraventricular beats (F1 about 0.2), because the RR intervals capture how early these beats arrive. The deep models do better on ventricular beats, and the beat-sequence context in the CNN-LSTM cuts down false alarms on normal beats. Still, no model reliably detects supraventricular or fusion beats in patients it has not seen, and macro-F1 stays below 0.40 across the board. Metaheuristic tuning raised validation scores but not test performance, since the validation beats came from the training patients. Under a strict inter-patient protocol, performance is held back more by the number of training patients and the class imbalance than by the choice of model family or tuner.

8 Future Work

- Switch to a patient-wise validation split (separate DS1 records) for both early stopping and the GWO/WOA fitness function, and rerun every model with at least three seeds.
- Feed the deep models RR-interval inputs normalised per patient, since the literature points to these as the key cue for S beats [12].
- Swap plain class weights for focal loss or batch-level resampling of the minority classes, to cut down false alarms on N beats.
- Bring in the MIT-BIH Supraventricular Arrhythmia Database or INCART to get more S and F examples, and more patients.
- Try an attention-based model as a fourth architecture family.

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Individual Contribution Statement

Credit for the project is shared equally across the team. The table below lists each member's main areas of work.

Member	Main contributions
Tanush Ralhan	1D-CNN and CNN-LSTM implementation, GWO/WOA tuning code, RR-interval features and SMOTE, final SVM/Random Forest run, error analysis and figures, final report
Pranshu Singh	Grid-search control experiment, Kaggle training runs for the 1D-CNN, grid-search, GWO and WOA tuned models and their results
Chinmai	Repository setup, dataset download script, initial preprocessing pipeline, first SVM/Random Forest baseline, synopsis, CNN-LSTM Kaggle training run

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by Pranshu Singh

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ECG-Based Arrhythmia Classification: A Comparative Study of Classical Machine Learning and Deep Learning Architectures Using Metaheuristic Hyperparameter Optimization

Deep Learning Mini Project – Phase 3 (Final Report)
[ICT-4442]

Submitted By

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SCHOOL OF COMPUTER ENGINEERING
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OCTOBER, 2026

Code repository: https://github.com/CHINMAYBHT/Arrhythmia-Detection

Abstract

Automatic heartbeat classifiction from the electrocardiogram (ECG) cojld make continuous carrdiac monitoring practical. Many published accuracies, however, come from train/test splits where beats from the same patient end up on both sides. In this project we put three model families through a single patient-independent protocol on the MIT-BIH Arrhythmia Database, using the DS1/DS2 record split and the five AAMI EC57 beat classes. The first is an SVM and a Random Forest trained on hand-crafted morphology, wavelet and RR-interval features. The second is a 1D convolutional neurral network (CNN) that reads the raw beat window. The third is a CNN-LSTM that looks at the five beats ending at the current one before majing its call. Alongside this, we checked whether two metaheuristic optimizers, the Grey Wolf Optimizer (GWO) and the Whale Optimization Algorithm (WOA), tune the CNN's hyperparameters better than a grid search given the same budget. On the 49,697 held-out test beats, the Random Forest came out on top inboth accuracy (0.860) and macro-F1 (0.393), with the SVM next (macro-F1 0.353). The best deep model, a WOA-tuned CNN, managed 0.346, and the CNN-LSTM 0.335. Every model picks up ventricular ectopic beats reasonably well (best F1 0.79). None of them reliably catches supraventricular (S) or fusion (F) beats, though, which keeps macro-F1 between 0.32 and 0.39. GWO and WOA did find configurations that scored higher on validation than anything the grid search found, but the adv-antage vanished on the test patients: the best tuned CNN (WOA) beat the untuned CNN on macro-F1 by just 0.005. Under a strict inter-patient protocol, then, the model family and the tuner matter much less than the small pool of training patients and the heavy class imbalance.

Keywords: ECG, arrhythmia classification, MIT-BIH, AAMI EC57, inter-patient evaluation, 1D-CNN, CNN-LSTM, Random Forest, SVM, Grey Wolf Optimizer, Whale Optimization Algorithm.

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1 Introduction

Cardiac arrhythmias are disturbances of the heart's rhythm, and some of them are early warnings of serious cardiovascular disease. Normally a cardiologist spots them byreading an ECG trace. That works, but it is slow, it varies from one reader to the next, and it cannot keep up with the hoyrs of signal that Holter monitors and wearable devices now record. Automatic beat classification has therefore been an active research topic for a long time, and the MIT-BIH Arrhythmia Database [1], [2] is the benchmark almost everyone reports on.

Broadly, the literature splits into two camps. Classical methods compute hand-crafted features for each beat (RR intervals, QRS morphology, wavelet coefficients and so on) and train a classifier such as an SVM or a Random Forest on top of them [12]. Deep learning methods skip the feature engineering and learn straight from the raw signal, using 1D CNNs [3], [13], [14] or recurrent and hybid CNN-LSTM networks that also model how one beat follows another [5], [7], [15]. Many of the headline accuracies above 95%, however, come from intra-patient splits, where the same patient's beats sit in both-the training and the test set. Switch to an inter-patient split, so that test patients are never seen during training, and performance on the minority classes falls-sharply [4], [5], [12], [16].

Our project takes one model from each camp and runs them all on the same data, with the same patient-wise split and the same metrics, so that whatever differences show up can be-put down to the models themselves. On the side, we ask whether two nature-inspired metaheuristic optimizers, GWO [9] and WOA [10], [17], can find better CNN hyperparameters than a grid search with the same compute budget. So far, that question has mostly been looked at for medical images [11].

2 Literature Review

The work most relevant to us falls into four groups: classical feature-based classifiers, convolutional networks, recurrent and hybrid networks, and-metaheuristic hyperparameter optimization. Table 1 summarises the main papers.

2.1 Evaluation protocol and classical feature-based methods

De Chazal, O'Dwyer and Reilly [12] introduceed the inter-patient DS1/DS2 split of MIT-BIH that we adopt here. They showed that a linear discriminant classifier on morphology and RR-interval features is highly sensitive to V beats in unseen patients but does much worse on S beats. Their work also made RR intervals essential for the S class, because a supraventricular beat looks al-most normal and differs mainly in arriving early. The five beat classes we use (N, S, V, F, Q) come from the AAMI EC57 standard [18]. Ben Slama et al. [8] found that an SVM trauned on CNN-extracted features can beat the CNN on its own (97.33% against 85.52%). A classical classifier given good features cantherefore outperform a deep network. Nikandish, He and Haghi [6] fused wavelet, interval and area features and lifted theaccuracy on difficult beat types considerably. Together these papers shaped Model 1: an SVM [19] and a Random Forest [20] on morphology, wavelet and RR-interval features. Finally, Silva et al. [16] reviewed studies from 2017 to 2024 and found that few of them follow the inter-patient protocol, which makes

published numbers hard to compare. That is part of why we hold every model-to one common protocol.

2.2 Convolutional networks

Kiranyaz, Ince and Gabbouj [13] showed that a compact 1D-CNN can-learn its own features from raw ECG beats, though their best results relied on patient-specific training. Acharya et al. [14] trained a 9-layer CNN on MIT-BIH beats and reported 94% accuracy, but on an intra-patient split. Khan et al. [3] reached 95.2% accuracy across five heartbbeat classes without saying whether their split kept patients apart. Guo, Sim and Matuszewski [4] tested DenseNet and GRU models on the inter-patient split andgot an F1 of 61.25 for S beats against 89.75 for V beats. Even for deep networks, then, S is the hardest class. Model 2 is a 1D-CNN alomg these lines.

2.3 Recurrent and hybrid networks

LSTM networks [15] are built-to capture long-range dependencies in sequences. Yildirim [7] paired a wavelet sequence with a bidirectional LSTM for ECG classification. Alamatsaz et al. [5] proposed a lightweight 11-layer CNN-LSTM with 98.24% mean accuracy across sveral databases. Mousavi, Afghah and Acharya [21] combined a CNN encoder with a sequence-to-sequence decoder; by modelling beat sequences and dealing-with class imbalance head-on, they reported a high inter-patient S-class sensitivity of 88.94%. Model 3 borrws that idea. A shared CNN encodes each beat, and an LSTM reads the sequence of five beats.

2.4 Metaheuristic hyperparameter optimization

The Grey Wolf Optimizer [9] imitates the leadership hierarchy and hunting behaviour of grey wolves, and the Whale Optimization Algorithm [10] imitates the bubble-net hunting of humpback whales; Mohammed, Umar and Rashid [17] surbey its many variants. Hamad, Samma and Suandi [11] reviewed metaheuristic CNN tuning for medical image di-agnosis between 2019 and 2022. GWO and WOA came up often, but almost every study dealt with images, and about 63.6% were about COVID-19 detection. We could not find any study that tunes an inter-patient ECG beat classifier with GWO or WOA and compares it against a grid search with an equal budget, so our side experiment does that.

Table 1: Summary of the main reviewed papers.

Ref	Author	Method	Dataset	Key result	Relevance
[12]	de Chazal et al. (2004)	LDA on morphology + RR features	MIT-BIH, inter-patient DS1/DS2	First standard inter-patient protocol; S class much harder than V	Our split; RR features (Model 1)
[4]	Guo et al. (2018)	DenseNet + GRU	MIT-BIH, inter-patient	F1 61.25 (S), 89.75 (V)	S is hardest even for deep nets
[3]	Khan et al. (2020)	CNN, 5 classes	MIT-BIH	95.2% accuracy (split not stated)	Reference for Model 2
[13]	Kiranyaz et al. (2016)	Compact 1D-CNN on raw beats	MIT-BIH	Learns features from raw ECG; patient-specific training	Basis of Model 2
[14]	Acharya et al. (2017)	9-layer CNN	MIT-BIH (intra-patient)	94% accuracy	Deep CNN baseline
[21]	Mousavi et al. (2018)	CNN + seq2seq	MIT-BIH, AAMI EC57	Inter-patient S sensitivity 88.94%	Beat sequences + imbalance (Model 3)
[5]	Alamatsaz et al. (2022)	Lightweight CNN-LSTM	MIT-BIH, LTAF	98.24% mean accuracy	Closest to Model 3
[7]	Yildirim (2018)	Wavelet sequence + Bi-LSTM	MIT-BIH	High accuracy on 5 rhythm classes	Recurrent ECG modelling
[6]	Nikandish et al. (2024)	Feature fusion + (Bi-)LSTM	MIT-BIH	RBBB 31.4% → 84.3% with area features	Wavelet / interval features
[8]	Ben Slama et al. (2025)	CNN features + SVM	DeepQ	CNN 85.52%, CNN-SVM 97.33%	Classical vs deep (Model 1)
[9], [10]	Mirjalili et al. (2014, 2016)	GWO, WOA	Benchmark functions	Competitive with PSO, GSA, DE	Our two optimizers
[11]	Hamad et al. (2024)	Review of metaheuristic CNN tuning	Medical images	GWO-tuned CNN 97.78% (COVID X-ray)	Motivation; no ECG studies
[16]	Silva et al. (2025)	Systematic review	ECG studies 2017–2024	Few use inter-patient splits	Justifies common protocol

3 Objectives

1. Build one shared, patient-independent preprocessing pipeline for MIT-BIH so that every model gets exactlythe same inputs and test data.
2. Implement a classical baseline (SVM / Random Forest), a 1D-CNN and a CNN-LSTM, compare them on the same metrics (accuracy, per-class precision/recall/F1, macro-F1, confusion matrix), and work out why they-perform differently.
3. Find out whether tuning the 1D-CNN's hyperparameters with GWO and WOA helps, compared with both the untuned net-work and a grid search on the same compute budget.

4 Methodology

4.1 Shared data pipeline

Every model draws on the MIT-BIH Arrhythmia Database [1]: 48 half-hour, two-channel ambulatory recordings-from 47 subjects, sampled at 360 Hz, with each beat annotated by cardiologists. We download the data with the `wfdb` library (`src/download_data.py`) and process it once with `src/preprocessing.py`:

1. **Filtering.** We band-pass filter channel 0 (MLII) betw-een 0.5 and 40 Hz using a third-order zero-phase Butterworth filter, which removes bsaeline wander and high-frequency noise.
2. **Segmentation.** Around every annotated beat we cut a 200-sample window: 90 samples beore the R-peak and 110 after it, or aobut 0.56 s.

3. **Labelling.** MIT-BIH symbols are mapped onto the AAMI EC57 classes [18]: N (N, L, R, e, j), S (A, a, J, S), V (V, E), F (F) and Q (/, f, Q). Anything that is not a beat annotatin is thfown away.
4. **RR intervals.** For each beat we coompute the previous RR interval, the next RR interval, the mean of the previous ten RR intervals and the ratio of previous to next RR. All of this happpens within the beat's own record, so no interval ever spans two recordings.
5. **Patient-independent split.** We follow de Chazal et al. [12]: 22 records make up the training-set (DS1) and a different 22 make up the test set (DS2). The four paced-records (102, 104, 107, 217) are left out. No patient sohws up in both sets.

That laves 51,008 training beats and 49,697 test beats (Table 2). The imbalance is severe. Roughly 89% of beats are Normal, and the Q class-has just 8 training beats and 7 test beats.

Table 2: Class distribution of the patient-independent split.

Class	Train beats	Train %	Test beats	Test %
N – Normal	45,854	89.9%	44,245	89.0%
S – Supraventricular ectopic	944	1.9%	1,837	3.7%
V – Ventricular ectopic	3,788	7.4%	3,220	6.5%
F – Fusion	414	0.8%	388	0.8%
Q – Unknown	8	0.0%	7	0.0%
Total	51,008	100%	49,697	100%

4.2 Model 1: SVM and Random Forest on hand-crafted features (Tanush Ralhan)

We describe each beat with 20 features. Eight are morpohlogy statistics of the window: maximum, minimum, mean, standard deviation, skewness, kurtosis, peak-to-peak amplitude andthe sum of absolute first differences. Another eight come from wavelets, namely the mean and standard deviation of each sub-band of a three-level Daubechies-4 decomposition. The last four are the RR-interval features. Standardisation usesstatistics from the training set only.

To deal with the imbalance, we oversample the training set with SMOTE [22], bringing each minority class-up to 20,000 samples (synthetic plus real) and leaving the N class as it is. The test set is nevver resampled. On top of th both classifiers use balanced class weights. For the Random Forest [20] we grid-search the number of trees (200, 400), the maximum depth (unlimited, 20) and the minimum leaf size (1, 2), using three-fold stratified cross-validation scored on macro-F1. The RBF-kernel SVM [19] is tuned over $C \in \{1, 10\}$ and $\gamma \in \{\text{scale}, 0.01\}$ on a 20,000-sample subset. The fianl SVM is then trained on a 60,000-sample subset of the oversampled data, since SVM training time grows faster than linearly with the number of samples. The search picked 400 trees, unlimited depth and a minimum leaf size of 1 for the Random Forest, and $C = 10$, $\gamma = \text{scale}$ for the SVM. Cross-validated macro-F1 on the oversampled training data came out at 0.997 and 0.984 respectively, far above the test results; Section 6 explainsthe gap.

4.3 Model 2: 1D-CNN on raw beats (Pranshu Singh)

Each 200-sample window is z-score normalised-per beat and passed to a 1D-CNN made of three blocks of Conv1D, batch normalisation, ReLU and max-pooling (32, 64 and 128 filters, kernel size 7, pool size 2). After that come dropout (0.3), global average pooling, a 64-unit dense layer and a five-way softmax, for 81,605 parameters in total. We train it with Adam [23] at a

learningrate of 10^{-3} and a batch size of 128, with balanced class weights in the cross-entropy loss, early stopping on validation loss (patience 8), learning-rate halving on plateau, and a cap of 50 epochs.

4.4 Model 2 tuning: grid search, GWO and WOA (Pranshu Singh)

All three tuners get exactly the-same budget: 36 short training runs (no more than 8 epochs each, early-stopping paitence 3). Every run is scored by macro-F1 on a validation set holding 15% of the training beats, drawn as a stratified random split; the test set plays no part in the search. Each tuner's best configuration is then retrained on the full training set for up to 50 epochs and evaluated on the test set once.

- **Grid search** covers the learning rate $\{3\times10^{-4}, 10^{-3}, 3\times10^{-3}\}$, the base filter count {16, 32, 64} (the blocks use base, 2× and 4× filters) and the kernel size {3, 5, 7, 9}, giving 3×3×4 = 36 points. Dropout (0.3) and dense uints (64) are held fixed.
- **GWO** [9] and **WOA** [10] search a continuous five-dimensional space: $\log_{10}$ learning rate in [-4, -2], dropout in [0.1, 0.5], base filters in [16, 128], dense units in [32, 256] and kernel sizein [3, 11] (rounded to an odd integer). Each optimizer-runs 6 agents for 6 iterations, which comes to 36 evaluations. In GWO, each wolf moves towards the average of the positions suggested by the three best wolves (alpha, beta and delta), while a coefficient a falls linearly from 2 to 0 so that the search gradually shifts from exploration to exploitation. In WOA, each whale picks one of three moves at random, with equal probability: it encircles the best whale, heads towards a random whale (exploration), or follows a logarithmic spiral around the best whale.

4.5 Model 3: CNN-LSTM hybrid (Chinmai)

The CNN-LSTM classifies each beat from the sequence of five consecutive-beats ending at it, which gives it rhythm context that a single-beat model never sees. Sequences are built inside each record (never across two recordings) with a stride of one beat. The firrst four beats of every record lack a full history and are dropped, which leaves 49,609 test sequences. A shared convolutional encoder (two blocks of Conv1D with 32 and 64 filters, kernel size 5, batch normalisation, ReLU and pooling, then global average pooling) is applied to every beat through TimeDistributed layers. An LSTM with 64 units [15] then reads the five beat vectors, followed by dropout (0.3), a 64-unit dense layer and a softmax over the five classes, for 48,389 parameters. Training uses the same optimiser, class weights and callbqcks as the CNN, but with a batch size of 64.

5 Experimental Setup

- **Hardware and software.** Model 1 ran on a laptop CPU with Python 3.14, scikit-learn and imbalanced-learn. TensorFlow has no stable release for Python 3.14 and we had nolocal GPU, so Models 2 and 3 and all of the tuning ran in Kaggle notebooks on an NVIDIA Tesla T4 GPU (`kaggle_pipeline.ipynb`, `kaggle_phase2.ipynb`). One epch took about 3 s for the CNN and about 18 s for the CNN-LSTM, and the grid search took 959 s. The final SVM/Random Forest run (feature extraction, SMOTE, both gridsearches and the final fits) took around 35 minutes on the CPU.

- **Metrics.** We repot accuracy, per-class precision, recall and F1, macro- and weighted-average F1, and theconfusion matrix (all from scikit-learn). Since N makes up 89% of the test set, macro-F1 is our main metric for comparison. Q has only 7 test beats, so we also give macro-F1 over N, S, V and F.
- **Common protocol.** Every model is trained on DS1 and evaluated once on DS2. Per-model results live in `results/`, and `src/compare_results.py` reb-uilds the compafison table, the error analysis and the figures from them.
- **Seeds.** Each configuration was run once, with the seed fixxed at 42.

6 Results and Discussion

6.1 Comparison across models

Table 3 puts all the models side by side on the-test set, and Figure 1 plots the per-class F1.

Table 3: Test-set results on the patient-independent DS2 split (single run each). Macro-F1* excludes the Q class (7 test beats).

Model	Acc.	Macro F1	Macro F1*	Wtd F1	F1 N	F1 S	F1 V	F1 F	F1 Q
SVM	0.777	0.353	0.442	0.828	0.872	0.208	0.678	0.009	0.000
Random Forest	**0.860**	**0.393**	**0.491**	0.881	0.924	0.192	0.785	0.064	0.000
1D-CNN (default)	0.727	0.342	0.426	0.800	0.839	0.085	0.779	0.002	0.004
1D-CNN (grid-search tuned)	0.703	0.339	0.422	0.783	0.819	0.090	0.777	0.004	0.003
1D-CNN (GWO tuned)	0.751	0.323	0.403	0.807	0.853	0.053	0.702	0.002	0.003
1D-CNN (WOA tuned)	0.763	0.346	0.432	0.824	0.865	0.070	0.794	0.002	0.003
CNN-LSTM	0.834	0.335	0.419	0.856	0.907	0.049	0.714	0.005	0.000

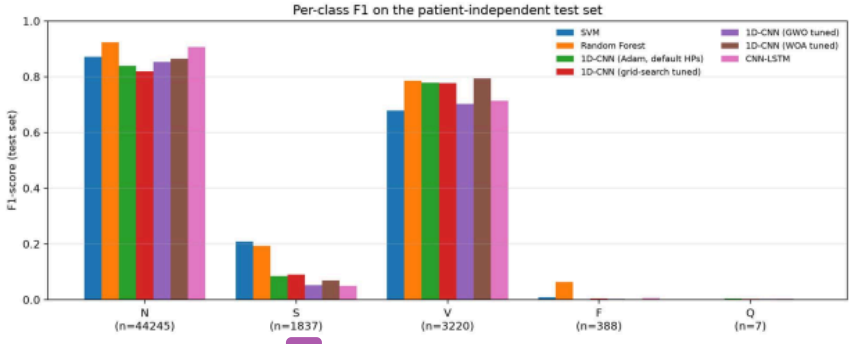


Figure 1: Per-class F1 of every model on the test set.

Of all the models, the Random Forest on hand-crafted features has the best macro-F1 (0.393, or 0.491 without Q), ah-ead of the best deep model, the WOA-tuned CNN at 0.346. It is the only model that picks up a useful share of S beats (F1 0.19, while no network gets past 0.09), and it also has the best F-class result (F1 0.06, recall 0.29). This comes from the RR-interval features: an S beat looks almost normal but comes early, and the previous-RR and RR-ratio features measure that timing. A networkthat only sees one 0.56 s window has no way to measure it. These

features were missing from the interom Random Forest run (macro-F1 0.31, S F1 0.00), so most of the imrpovement since then comes from them and from SMOTE. The SVM, which uses the same features, has the highest S-class F1 (0.21) and the second-best macro-F1 (0.353). However, it labels 5,969 Normal beats as Fusion, and that drags its accuracy down to 0.777. Both classical models also overfit in the same way as the tuning experiment described below. Their cross-validated macro-F1 on the training data was 0.98–1.00, compared with 0.35–0.39 on the test patients. This happens bec-ause the cross-validation folds contain beats from the same patients, and SMOTE builds synthetic beats from neighbours that end up in different folds.

The deep models learn the shape of ventricular beats well with no hand-crafted features at all: the CNN variants reach a V-class F1 of 0.70–0.79, on par with-the Random Forest (0.79). A ventricular beat has a wide, abnormally shaped QRS complex, and convolutional filters pick that up straight from the waveform. Among the deep models, the CNN-LSTM has the highest accuracy (0.834) and the fewest false al-arms on Normal beats (N recall 0.871, against 0.758 for the CNN). Seeing the four preceding beats helps it confirm that a beat is part of a regular normal rhythm. Even so, its macro-F1 (0.335) is no better than that of the single-beat CNN (0.342). What it gains on N it gives back elsewhere, with more Normal beats wrongly labelled S (3,871) and a lower V precision (0.62). The LSTM cznnot learn RR timing from the beat sequence either, because each input window iscentred on its own R-peak, so the gaps between beats never show up in the sequence of windows; only the shapes of the neighbouring beats do.

6.2 Error analysis

Figure 2 shows the row-normalised confusion-matrices (the raw counts are in `results/error_analysis.txt`). Three error patterns turn up in every model.

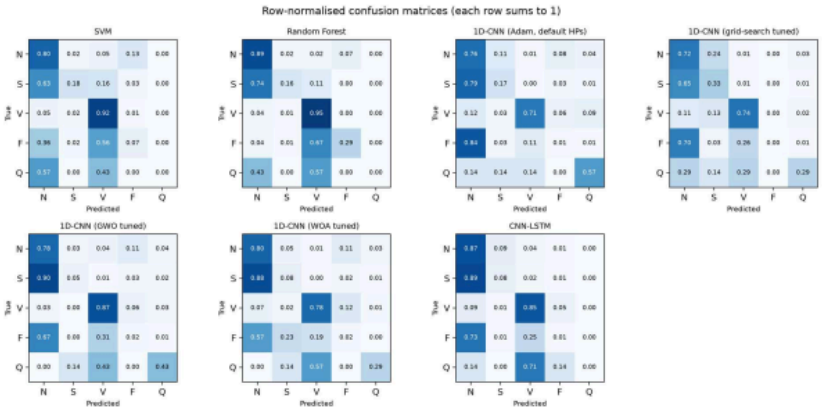


Figure 2: Row-normalised confusion matrices (rows are true classes, each row sums to 1).

1. **S beats are classified as Normal.** Every model predicts most true S beats as N, from 63% for the SVM up to 90% for the GWO-tuned CNN. A supraventricular ectopic beat starts above the ventricles, so its QRS complex looks almost exactly like a normal one; what sets it apart is that it arrives early. The networks only ever see R-peak-centred windows, so this timing is invisible to them, and they miss 65–90% of S beats. The SVM and Random Forest

get the RR ihtervals and miss the-fewest (63% and 74%). Even with timing information, though, they still miss most S beats, because whether a beat is premature has to be judged against the patient's own heart rate, and the training set holds only 944 S beats. The S class is-also concentrated in very few patients. Of the 1,837 test S beats, 1,382 (75%) come from a single record (232), and 383 of the 944 training S beats come from record 209. In effect, the modelslearn the S beats of a handufl of training patients and are then tested mostly on one different patient whose beats they fail to generalise to.

2. **Class weights cause many false alarms on Normal beats.** With balanced class weights, a sin-gle Q training beat counts roughly 1,275 times as much as an N beat. The networks respond by pushing part of the huge N class into the rare classes. The untuned CNN, for example, labels 5,013 N beats as S, 3,659 as F and 1,733 as Q, and togetuer these account for 77% of its mistakes. Because N is so large, gettingeven a few percent of N beats wrong is enough to push the precision of S, F and Q close to zero.
3. **Fusion beats are confused with V and N.** A fusion-beat is a blend of a normal and a ventricular beat, so its shape fa-lls somewhere between the two. The classical models call most F beats V (56–67%), whereas the networks call most of them N (57–84%), and F-class F1 is at most 0.013 for every model. It does not help that 372 of the 414 training F beats come from one patient (record 208) and 362 of the 388 test F beats from another (record 213), so the F class is learnd from one patient and tested on a different one.

The Q-class scores mean very little. With only 7 test beats, a single correct prediction moves Q reecall by 0.14.

6.3 Metaheuristic versus grid-search tuning

Table 4 and Figure 3 set-the four CNN variants against each other.

Table 4: 1D-CNN tuning experiment. All tuners had the same budget of 36 short training runs.

Variant	Val. macro-F1	Test acc.	Test macro-F1	Selected hyperparameters
Default (Adam)	–	0.727	0.342	lr 10^{-3}, filters 32/64/128, kernel 7, dropout 0.3, dense 64
Grid search	0.462	0.703	0.339	lr 10^{-3}, filters 32/64/128, kernel 7, dropout 0.3, dense 64
GWO	0.627	0.751	0.323	lr 5.5×10^{-4}, filters 78/156/312, kernel 11, dropout 0.16, dense 161
WOA	0.556	0.763	0.346	lr 1.3×10^{-4}, filters 125/250/500, kernel 7, dropout 0.17, dense 75

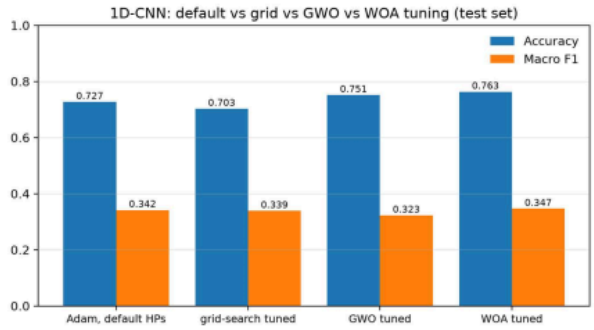


Figure 3: Test accuracy and macro-F1 of the four CNN variants.

Both GWO and WOA found configurations with clearlyhigher validation macro-F1 than the best grid piont (0.627 and 0.556, against 0.462). That is because their continuous search space also tkes in dropout, the dense layer size and larger networks, all of which the grid kept fixed. None of this carried over to the test set. WOA lifted test macro-F1 by only about 0.005 over the untuned CNN, and GWO was actually 0.019 worse, even though both raised accuracy by 2–4 points simply by predicting fewer S beats. The grid search ended up selecting exactly the default configuration, yet its retrained model scored 0.024 lower in accuracy than the default run. That alone shows how much a single training run can vary.

The cause is the-validation set used during the search. It is a random 15% of the training beats, so it contains beats from the same patients as the training data, while the test set is made up of new patients. The optimizers therefore end up choosing hyperparameters that suit the training patients. Both picked larger networks than the default (78/156/312 filters for GWO and 125/250/500 for WOA, against 32/64/128) and lower dropout, around 0.16-0.17 instead of 0.3. This is the kind of overfitting the inter-patient protocol is meant to expose. On the test set, the spread between configurations (0.323–0.346) is smaller-than the variation between two runs of the same configuration. On this dataset, metaheuristic tuning did not pay for itself: each optimizer cost 36 extra training runs and delivered noreliable improvement. A fair test of these optimizers wouldneed a validation set drawn from separate training patients.

6.4 Comparison with published work

Our scores fall well-short of the 94–98% accuracies reported in [3], [5], [14], but those studies eitehr use intra-patient splits or do not say which split they used. Under the inter-patient protocol, published S-class F1 values sit around 0.5–0.6 [4], [12], while our models reach 0.09 at best. Most of that gap comes down to techniques we did not use: per-patient RR-interval normalisation [12], sequence-to-sequence modelling with targeted resampling [21], or larger models trained on additional databases [4].

6.5 Limitations

1. **Early stopping on the test set.** The CNN, its tuned retrains and the CNN-LSTM all used the test set asthe validation set for early-stopping and learning-rate reduction, so their test scores are slightly optimistic. The SVM and Random Forest are unaffected. Fixing this

means holding out separate training patients for validation and rerunning the networks on a GPU.

2. **Single run-per model.** Differences of a few points between mdels fall within the run-to-run variation (the default CNN, for instance, gave 0.727 and 0.733 accuracy over two runs), so they should not be treated as significant.
3. **The CNN-LSTM is scored on 88 fewer test beats** (the first four beats of each record), so its test set is not quite the same as the ot-her models'.
4. **Q class.** With only 8 training and 7 test beats, Q can be neit-her learned nor evaluated, wh-ich is why we also report macro-F1 without it.

7 Conclusion

We compared an SVM, a Random Forest, a 1D-CNN and a CNN-LSTM for five-class ECG beat classification on MIT-BIH, all under the-same patient-independent protocol, and we tested GWO and WOA againt a grid search for tuning the CNN. The Random Forest, with RR-interval features and SMOTE, gave the best accuracy (0.860) and macro-F1 (0.393). The two classical models were also the only ones to catch a meaningful share of supraventricular beats (F1 ab-out 0.2), because the RR intervals capture how early these beats arrive. The deep models do better on ventricular beats, and the beat-sequence context in the CNN-LSTM cuts down false alarms on normal beats. Still, no model reliably detects supraventricular or fusion beats in patients it has not seen, and macro-F1 stays below 0.40 across the board. Metaheuristic tuning raised validation scores but not test performance, since the validation beats came from the training patients. Under a strict inter-patient protocol, performance is held back more by the number of training patients and the class imbala-nce than by the choice of model family or tuner.

8 Future Work

- Switch to a patient-wise validation split (separate DS1 records) for both early stopping and the GWO/WOA fitness function, and rerun every model with at least thrree seeds.
- Feed the deepmodels RR-interval inputs normalised per patient, since the literature points to these as the key cue for S beats [12].
- Swap plain class weights for focal loss or batch-level resampilng of the minority classes, to cut do[29]false alarms on N beats.
- Bring in the MIT-BIH Supraventricular Arrhythmia Database or INCART to get more S and F examples, and-more patients.
- Try an attention-based mo-del as a fourth architecture family.

Acknowledgement

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Individual Contribution Statement

Credit for the project is sahred equally across the team. The table below lists each member's main areas of work.

Member	Main contributions
Tanush Ralhan	1D-CNN and CNN-LSTM implementation, GWO/WOA tuning code, RR-interval features and SMOTE, final SVM/Random Forest run, error analysis and figures, final report
Pranshu Singh	Grid-search control experiment, Kaggle training runs for the 1D-CNN, grid-search, GWO and WOA tuned models and their results
Chinmai	Repository setup, dataset download script, initial preprocessing pipeline, first SVM/Random Forest baseline, synopsis, CNN-LSTM Kaggle training run

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